

Displacement and gentrification in England and Wales: A quasi-experimental approach

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Abstract

Using event history methods we examine the relationship between residential mobility and gentrification in England and Wales. While our models of residential mobility conform to the life-cycle explanations of mobility, we find little evidence of elevated mobility in gentrifying neighbourhoods. This finding is robust across a variety of specifications. We conclude by discussing possible explanations for this counter-intuitive result.

Keywords

displacement, England, event history methods, gentrification, residential mobility

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Introduction

This study uses quasi-experimental methods to examine the relationship between residential mobility and gentrification as a way of inferring the extent to which gentrification causes displacement. This approach is not without shortcomings, but it does shed light on the magnitude of an important potential outcome of gentrification – displacement.¹ To date much of the work on gentrification and displacement utilising this approach has been limited to the US. As the context of

residential mobility in the US differs substantially from other countries, findings based on US data may not be generalisable elsewhere. Our aim is to discern whether the US-based patterns of findings on gentrification and displacement utilising quasi-experimental methods holds in a different context – England and Wales.

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Previous findings

Randomised experiments are often considered the 'gold standard' for identifying causal relationships, but randomly assigning households to the 'intervention' of gentrification is both impractical and inappropriate for studying such a contested and dynamic process. Thus, the use of quasi-experimental methods that include a counter-factual is a realistic option for measuring the relationship between gentrification and displacement. Early and a few contemporary (Van Criekingen, 2009) studies of displacement using succession and retrospective methodologies failed to include such a counter-factual (see Freeman, 2005 for a detailed discussion of these approaches).

After these early displacement studies, transatlantic debates over displacement explored whether the working class and poor were being replaced rather than displaced in areas affected by long-term decline and/or the construction of new developments. Although the term gentrification was first coined in England and much theorising on its causes and extent took place here, relative to the US, there was a paucity of empirical studies on displacement. In an important advance Atkinson (2000) used longitudinal individual-level data to examine the migration of gentrifiers and potential displacees, respectively, into and out of gentrifying neighbourhoods. Atkinson found high rates of net outmigration from gentrifying areas among those who were inactive, the working class and the elderly, consistent with both direct displacement and exclusionary displacement. Because Atkinson did not measure mobility directly, however, we cannot as Atkinson concedes attribute the observed net migration patterns to direct or indirect displacement. Moreover, Atkinson's failure to control for other factors that contribute to residential mobility also limits our ability to infer whether the elevated

migration rates among some groups in his study were due to gentrification or individual characteristics associated with mobility, such as the stage of the life cycle.

Several scholars focusing on the UK also provided a theoretical justification for how gentrification might occur without causing displacement. Hamnett and Whitelegg (2007: 122) describe gentrification occurring in London without having '... been accompanied by significant residential displacement as almost all the new housing units were in what were previously warehouses, industrial, or office buildings.' Elsewhere Hamnett (Butler and Hamnett, 2009; Hamnett, 2003) describes how the professional classes in the UK are comprising an ever larger share of the population and are coming to replace, more so than displace, the working classes. Hamnett argues that as the working class shrinks and the professional class grows, it is to be expected that some of the neighbourhoods inhabited by the working class would gentrify. This could occur without necessarily directly displacing poorer, working class households. While plausible, to our knowledge Hamnett's hypothesis has never been formally tested.

In the first decade of this century American scholars began to take advantage of increasingly available longitudinal micro-data to measure the relationship between gentrification and displacement using quasi-experimental methods. The key contributions of this approach are the comparison of mobility or displacement levels in gentrifying and non-gentrifying areas, and the inclusion of measures that control for characteristics other than gentrification (such as life cycle and housing characteristics) that influence mobility. Taken together, these features allow us to infer how much of the observed migration or displacement from gentrifying neighbourhoods can be attributed to gentrification.

The evidence on gentrification and displacement from this body of work is mixed.

Vigdor (2002) analysed American Housing Survey (AHS) data to measure displacement from gentrifying zones of Boston between 1985 and 1993. Zones were classified as gentrifying based on the relative increase in the college-educated share of the population; the study also tested a more restrictive definition consisting of one central zone of the city. Using either definition, Vigdor found no evidence that households headed by someone without a postsecondary education were more likely to leave their units in gentrifying zones than elsewhere in the metropolitan area. Similarly, Freeman and Braconi (2004) analysed New York City Housing and Vacancy Survey (NYCHVS) data from the period 1991–1999 to examine displacement from seven gentrifying sub-borough areas (SBAs). The classification was based on the authors' knowledge of local neighbourhood change and supported by data documenting demographic and socioeconomic shifts consistent with gentrification. They found that poor households and those without a four-year college education living in gentrifying SBAs were less likely to move compared to similar households elsewhere in the city.

Subsequent studies analysed more fine-grained geographical units, limited the comparison group to low-income neighbourhoods that were similar but for experiencing gentrification, and increased the generalisability of their findings by using nationally representative datasets. For example, Freeman's (2005) study of displacement from the mid-1980s through the 1990s identified US census tracts that were potential candidates for gentrification based on their central city location, median income relative to the metropolitan area, and housing stock age. Using the Panel Study of Income Dynamics (PSID), he compared the likelihood of out-migration and displacement among residents of tracts that gentrified during the ensuing decade to residents of non-gentrifying tracts. When using an explicit

measure of involuntary mobility (rather than overall mobility), this study did find higher rates of displacement in gentrifying neighbourhoods, but this represented only a 1.4% probability of being displaced in gentrifying neighbourhoods versus a 0.9% probability in non-gentrifying neighbourhoods (Freeman, 2005). Moreover, Freeman's study failed to find consistently higher rates of mobility in gentrifying neighbourhoods.

A subsequent study of US Decennial Census data compared demographic shifts in low-income census tracts that experienced large increases in average family income between 1990 and 2000 to comparable tracts that did not experience such an increase. The authors did not observe significantly higher out-migration rates of households with low education levels, whatever their racial background, from low-income neighbourhoods classified as gentrifying (McKinnish et al., 2010). A similar analysis of the AHS from 1991 to 1999 found no evidence of higher mobility rates among US renters or poor households in low-income census tracts that experienced large gains in average household income relative to the metropolitan area compared to similar households in non-gaining tracts (Ellen and O'Regan, 2011). Using a similar operationalisation of gentrification as the two aforementioned studies, a study of PSID data from 1972 to 2003 found no consistent relationship between residence in a gentrifying tract and residential mobility (Lee, 2014).

Due to data constraints and the lack of conceptual consensus over what constitutes 'gentrification', 'displacement' or even 'neighbourhood', the constructs used in these studies are necessarily approximations of the phenomena of interest. For example, while census tracts are closer proxies for neighbourhoods than the larger-scale units analysed by Vigdor (2002) and Freeman and Braconi (2004), their boundaries may still not be coterminous with those that residents

identify with a particular neighbourhood. Moreover, gentrification is a dynamic process that can occur gradually, unevenly, and over long periods of time; as discussed below, its effects on displacement may vary at different stages. This underscores the sensitivity of results to the time frame chosen (and available) for analysis. Moreover, operationalisations of gentrification vary. While some studies focus on rental inflation or tract-level income gains, others examine more explicit indicators of class such as increases in professionalisation or college-educated population. The latter choice is often based on the observation that early stages of gentrification are characterised by influxes of college-educated but relatively low-paid professionals (see Freeman, 2005 for a more detailed discussion of these issues).

One criticism of the aforementioned studies is their failure to capture households who move out of the study area or who become homeless, as Newman and Wyly (2006) found happens to residents in gentrifying neighbourhoods in New York City. This criticism is unfounded, however, as a household that was displaced and became homeless, doubled up or left the city would still be captured as having vacated their previous unit in such studies.

The notion that displacement is experienced in ways that are not captured by statistical surveys has also been advanced by studies of new-build gentrification in London, which describe how the immigration of higher-income households impacts residents of adjacent neighbourhoods through increased insecurity about housing prices and changes in neighbourhood culture and amenities (Davidson, 2008; Davidson and Lees, 2005, 2010). These studies provide an important reminder that gentrification affects long-time residents even when, as is typical of new-build gentrification, reinvestment does not lead to

direct displacement. However, this does not discount the utility of longitudinal micro-data for measuring direct displacement where it does occur.

Another argument levelled by Slater (2009: 305), is that ‘... low mobility is to be expected’ in gentrifying neighbourhoods, because rising prices make it difficult for low-income households to find housing elsewhere. Slater argues that it is in non-gentrifying poor areas where the rent gap is highest and the potential profit from displacing poor residents greatest; consequently, mobility in these areas will be high as landlords disinvest and evict tenants in anticipation of incipient gentrification. A related argument holds more simply that including very poor neighbourhoods in the control group of non-gentrifying neighbourhoods sets a misleadingly high baseline for mobility because very poor renters move more frequently (Newman and Wyly, 2006). However, these depictions of low mobility rates in gentrifying neighbourhoods and elevated mobility rates in non-gentrifying, poor neighbourhoods appear inconsistent with Ellen and O’Regan (2011: 93), McKinnish et al. (2010) and Freeman (2005) finding no statistically significant differences in mobility between gentrifying and non-gentrifying low-income neighbourhoods once other controls for mobility are included.

A similar critique notes that displacement rates are likely to be higher during the early stages of gentrification, when those without the means to stay are pushed out. Consequently, studies that omit this time period will underestimate the incidence of gentrification-induced displacement. There is evidence that low-income households who remain in gentrifying neighbourhoods after the initial waves of reinvestment are able to do so due to subsidies, rent regulations or personal coping strategies (such as enduring a higher rent burden) that are not realistic

options for those who left (Newman and Wyly, 2006). As suggested in Slater's (2009) critique discussed above, those who remain in gentrifying neighbourhoods may also do so because of a lack of accessible or affordable options elsewhere (DeVerteuil, 2011, 2012; Newman and Wyly, 2006; Wyly et al., 2010). A study of social service providers in gentrifying neighbourhoods in Los Angeles and London (DeVerteuil, 2011, 2012) found that a fifth of interviewed facilities experienced direct displacement, while over half remained in place because of a lack of alternatives elsewhere. The absence of a comparison with non-gentrifying neighbourhoods precludes an estimate of how much of this experience is unique to gentrifying areas. Nonetheless, these findings highlight the importance of capturing the early stages of gentrification in analyses of displacement. As several of the above-mentioned quantitative studies compare gentrifying and non-gentrifying neighbourhoods that were similarly disadvantaged at the beginning of the study period, they are likely to capture these early stages of gentrification, particularly since they test multiple specifications of gentrification.

Despite an expanding understanding of the multiple facets of displacement and the need for a variety of methodologies to understand them, the incidence of direct displacement remains a prominent concern in gentrification debates and longitudinal microdata continues to provide a unique tool for understanding this relationship. Because mobility rates are substantially higher in the US than in other advanced industrialised countries (David et al., 2008; Schachter, 2004; Seko and Sumita, 2007), the generalisability of the aforementioned studies beyond the US is questionable. Relatively high mobility rates in the US may make it difficult to distinguish the 'signal' of displacement from the 'noise' of normal residential turnover. In a setting where mobility

rates are lower, such as England and Wales, displacement should be easier to discern. Despite being the home of the term gentrification and providing the raw material for some of the fiercest debates on the topic, the peer reviewed literature is devoid of quasi-experimental methods for detecting displacement in gentrifying neighbourhoods in England and Wales. This research tests whether the findings of US-based quasi-experimental studies of displacement in gentrifying neighbourhoods are robust across the Atlantic.

Data

Data for this study were drawn from the British Household Panel Survey (BHPS) and the UK census. The BHPS is an Economic and Social Research Council funded panel survey that began in 1991 with a nationally representative sample of approximately 5500 households (ISER, 2014). Individuals are followed over time and new individuals are added to the sample if they join a household that was in the original sample (Taylor et al., 2010). The BHPS collects a plethora of attitudinal, demographic, economic and social data from each participating household. Two features are particularly useful for this study. First, each year the BHPS asks respondents if they moved in the previous year. Second, after meeting certain security provisions, researchers can access geocoded versions of the BHPS that can be linked to the UK decennial census.

This study uses the 1991 through 2009 waves of the geocoded BHPS. After 2009 the BHPS was supplanted by the Understanding Society Panel Survey. As the first (1991) wave of the BHPS asked respondents how long they were living at their current address, we can infer how long the respondent has resided there in any given year by combining the length of time the respondent was at that address in 1991 with the observed durations

after 1991. For individuals who move after 1991 we can observe directly how long they resided at that address.

The 2001 through 2009 waves are linked to Lower Layer Super Output Area (LLSOA) data from the UK census to discern how neighbourhood characteristics are related to residential mobility. LLSOAs average 1500 persons and 650 households (Office for National Statistics, 2014). Because LLSOAs were first created in 2001, the analysis focuses on the 2001–2009 waves of the BHPS, only utilising data from earlier waves to calculate individuals' length of residence at their current location. As the BHPS asks respondents each year if they moved the previous year, we can infer what the respondents' neighbourhood conditions (e.g. gentrifying or not) were like in the previous year. This study focuses on England and Wales as those are the only two British countries that use the same small geographic areas – LLSOAs – to disseminate neighbourhood level data.

Analytic strategy

Our approach for discerning the relationship between gentrification and displacement closely follows the method employed by several US-based studies (Ellen and O'Regan, 2011; Freeman, 2005; Freeman and Braconi, 2004; Vigdor, 2002). The demographic composition of gentrifying neighbourhoods changes because of alterations in the flows of in-movers and out-movers. Displacement implies that at least some of the alteration in out-migration from the neighbourhood is involuntary. Although the BHPS asks respondents why they moved, the responses typically given do not map neatly onto the notion of displacement. For example, rising rent or landlord harassment are not among the responses listed. In addition, retrospective responses might not always align with our notions of displacement. For example, if

a landlord in a gentrifying neighbourhood raised a tenant's rent, the tenant might decide this is the impetus for moving in with a romantic partner or seeking a job in another city. While the rent increase motivated a major life choice that resulted in a move, if asked why they moved, the respondent might say 'to move in with their partner' or 'for a job', respectively.

For these reasons, we infer displacement indirectly by testing whether mobility rates are higher in gentrifying neighbourhoods even after controlling for household level factors that have been found to predict mobility (Cho and Whitehead, 2013; Dieleman, 2001; Owen and Green, 1992; Rossi, 1980).

We use event history methods to analyse residential mobility in gentrifying and non-gentrifying neighbourhoods. The 'event' of interest is whether someone moves. We are interested in how the neighbourhood conditions (i.e. undergoing gentrification or not) at a specific point in time are related to the probability of moving at that time. While we will observe some moves by individuals over the course of the study period, some households will be in the midst of a residential spell whose termination will not be observed (i.e. right censored-spells). Event history methods allow us to include these spells in the analysis without arbitrarily assuming these individuals will never move. Because the BHPS collects data at discrete, annual intervals, we employ a discrete time logistic regression model. We use logistic regression because the outcome, to move or not, is a binary outcome that can be modelled:

$$\log\left(\frac{P_{it}}{1 - P_{it}}\right) = \alpha_t + \beta_1 x_{it1} + \beta_2 w_{it2} \dots + \beta_k x_{ikt}$$

where P_{it} is the probability of moving at time t for an individual with unvarying covariates X_{it} (e.g. gender) and time-varying covariates

W_{it} (e.g. age). B_1 and B_2 are unknown parameters. In our discrete time framework the data are structured in person-year format where each individual is represented in the data equal to the number of years they were part of the BHPS. Having multiple observations from the same individual violates one of the assumptions made when conducting tests of statistical significance – independent observations. We estimate robust standard errors to be able to relax this assumption.

LLSOAs serve as proxies for neighbourhoods in this analysis. Gentrification is conceptualised here as a class-based phenomenon whereby lower-status neighbourhoods change to higher-status ones. Following Atkinson (2000) and Hamnett (2003) we base our measure of class and consequently gentrification on the degree of professionalisation or white-collar positions in a neighbourhood. The ‘gentry’ are those in the following National Statistics Socio-economic Classification (NS-SEC) strata: higher managerial, administrative and professional occupations; large employers and higher managerial and administrative occupations, and lower managerial, administrative and professional occupations. As the UK deindustrialises, more people are employed in these occupations. The average representation in LLSOAs in these professional strata increased from 25.9% to 30.2% between 2001 and 2011. Our measure of gentrification is hence a relative one. Two criteria were used to define a LLSOA as gentrifying. First, in 2001, the LLSOA had to have a representation of professionals that was below the median for all LLSOAs. This is our measure of relative disadvantage. Second, a LLSOA had to experience an increase in professionalisation between 2001 and 2011 that was at the 75th percentile or above for all LLSOAs. Thus, relatively disadvantaged LLSOAs that experienced an above average increase in professionalisation were classified as gentrifying.

We perform two sets of analyses: one for all LLSOAs in England and Wales and a second for all LLSOAs in the Greater London region. London is a world city, provided the inspiration for the original conception of the word gentrification, and has been the site of extensive gentrification (Hamnett, 2009) and thus deserves special consideration.²

Relatively disadvantaged LLSOAs that did not experience gentrification (as defined here) serve as our control neighbourhoods. We compare the experiences of those residing in gentrifying neighbourhoods with those residing in disadvantaged neighbourhoods that did not gentrify and those residing in advantaged neighbourhoods (neighbourhoods in the top half of the distribution in terms of the gentry in 2001). Table 1 illustrates the characteristics of these three types of neighbourhoods along the dimensions of number of gentrifying households, the number of households experiencing four measures of deprivation (unemployment, low educational attainment, poor health and inadequate housing), the number of households experiencing no deprivation measures, and the proportion of housing that is social housing, for England and Wales and for London, respectively. The advantaged neighbourhoods have fewer deprived households and less social housing over the entire study period. The gentrifying and non-gentrifying neighbourhoods start out roughly comparable but the gentrifying neighbourhoods experience a sharper increase in the number of gentrifiers and a sharper decline in deprivation and social housing than the non-gentrifying neighbourhoods. This pattern is starker for London than for the entire sample. These differences were statistically significant. This is consistent with a view of gentrification as a process of remaking poorer working-class neighbourhoods into more affluent spaces.

This process can be illustrated by examining some of the gentrifying areas shown in

Table 1. Characteristics of neighbourhoods.

	Gentry households		Four deprivation households		Zero deprivation households		Social housing	
	2001	2011	2001	2011	2001	2011	2001	2011
Non-gentrifying	201	235	9.9	4.7	145.1	216.5	26.1%	24.5%
Gentrifying	174.5	283.2	11.2	5.1	133.2	252.8	30.7%	26.9%
Advantaged	400	476	3.6	2.3	245	335.6	10%	10%
London								
Non-gentrifying	2001	2011	2001	2011	2001	2011	2001	2011
Gentrifying	225.11	239.4	9.8	7.7	133.3	162.2	39.9%	38.1%
Advantaged	203.8	311.6	13.3	10.4	119.4	181	51.2%	46%
Advantaged	451.9	535.2	6.4	5.1	247.6	304.9	18.1%	17%

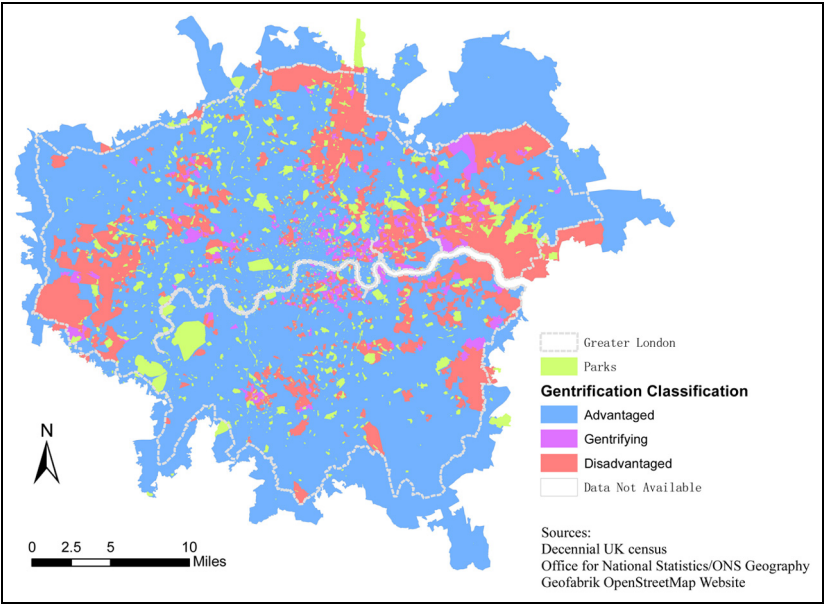


Figure 1. Classification of Greater London neighbourhoods.

Figure 1, which maps this measure of gentrification for the Greater London Area. Swathes of LLSOAs in traditionally working-class areas of East London experienced gentrification between 2001 and 2011, including in relatively deprived riverside boroughs like Tower Hamlets, Newham and Greenwich that have experienced waves of redevelopment since the decline of the docks

(Fainstein, 2010; Leeser, 2010; Watt, 2013). Large-scale regeneration plans, such as those associated with London Docklands and the 2012 Olympics, have aimed to transform the economic structure, physical accessibility, socioeconomic mix, and overall image of affected East London boroughs (Brownhill, 1999; Fainstein, 2010; Host Boroughs, 2009; Mayor of London, 2007). Since the 1990s

this vision has been implemented, *inter alia*, through the siting of Canary Wharf financial centre on former portlands; the construction of high-density residential developments oriented to affluent professionals on vacant riverside land next to low-income neighbourhoods; major investment in transportation infrastructure; large-scale commercial, residential and recreational development on brownfield and former industrial sites (in some cases involving the removal of existing manufacturing enterprises and low-cost housing); and the mixed-tenure redevelopment of council housing estates (Brownhill, 1999; David and Thornley, 2012; Davidson and Lees, 2005, 2010; Watt, 2013). The identity rebranding and cultural shifts associated with such redevelopment have elicited concern, resistance and reports of indirect displacement from ethnic minority and working-class residents, and amid rising housing prices and high levels of housing need in East London, reports and fears of direct displacement are also common (Davidson, 2008; Davidson and Lees, 2005; Watt, 2009, 2013).

To assess whether gentrifying neighbourhoods experience significantly elevated rates of direct displacement, this study models residential mobility to observe if there is a surfeit of mobility in these neighbourhoods. Using the notion of a life-cycle to inform our model, we include measures for age, marital status, gender and the presence of children. Tenure has also been found to predict mobility, with owners substantially less likely to move (Rohe and Stewart, 1996). In the UK, residence in social housing has been associated with lower rates of mobility (Cho and Whitehead, 2013). Therefore both tenure and the type of rental are included as control variables. Finally, we use fixed effects to account for the respondent's local authority, which addresses local differences in housing context and problems that might arise from spatial autocorrelation.

Because gentrification is a process that purportedly drives disadvantaged persons from their neighbourhoods, we examine whether low-income or working-class individuals are particularly vulnerable to higher rates of gentrification-induced mobility by including an interactive term between residence in a gentrifying LLSOA and being low-income or working-class, respectively. Low-income individuals were defined as those in the bottom quintile of the income distribution in a given year. Working-class individuals were defined as those whose latest job was in one of the following occupations: personal service work, foreman manual, skilled manual workers, semi-skilled manual workers, unskilled manual workers, own account workers, agricultural workers and members of armed forces. Finally, we also model the hazard rate, or the probability that an individual will move in a given year.

Table 2 illustrates the frequencies of the control variables in our regression model. Approximately 8% (9% in London) of the sample move in a given year. This figure, at least 50% lower than estimates of residential mobility in the US (Freeman, 2005; Schachter, 2004), confirms that England and Wales are an appropriate context to test whether lower rates of residential mobility will make it easier to detect the signal of displacement from the noise of residential turnover.

In sum, our approach is to use event history methods to model how likely individuals are to move in gentrifying neighbourhoods compared to other disadvantaged neighbourhoods that do not gentrify using life-cycle and housing context factors as statistical controls. We use the presence or extent of gentrification as a predictor of residential mobility in that year. We also test whether low-income and working-class individuals are especially vulnerable to displacement in gentrifying LLSOAs.

Table 2. Descriptive statistics used in regression models.

	Full sample	London only
Moved	7.8%	9.1%
Has a child	33.2%	28.3%
Aged 18–24	13.3%	11.3%
Aged 35–44	18.9%	19.5%
Aged 45–54	16.2%	15.5%
Aged 55–64	14.6%	14.9%
Aged 65 and above	20%	18.3%
Male	45.9%	45.4%
Divorced, separated or widowed	17.6%	15.4%
Never married	29.7%	39.9%
UK citizen	96.4%	90.6%
Council resident	10.2%	12%
Property company	1%	1.7%
Other association	4.6%	6.3%
Other renter	7.6%	10.5%
Urban	68.2%	
Duration 3–5 years	23.3%	19.3%
Duration 6–10 years	18.5%	15.4%
Duration greater than 10 years	26.3%	35.8%
Low income	18.2%	15.2%

Results

In the interest of brevity we present the full regression model for only two specifications and discuss the full results of only one model. Table 3 illustrates the results of four discrete time logistic regression models, the first two columns for the entire sample and the last two columns for London. The first and third columns have as their key independent variable an interaction term between low-income status and residence in a gentrifying, non-gentrifying or advantaged neighbourhood. The second and fourth columns have as their key independent variable an interaction term between working-class status and residence in a gentrifying, non-gentrifying or advantaged neighbourhood. Significance tests are at the 95% level of confidence and are based on robust standard

errors adjusted to account for clustering among observations. The regression coefficients are presented as odds ratios. Values greater than one indicate a positive relationship with the probability of moving and values less than one indicate a negative relationship.

The results in Table 3 mostly conform to the tenets of life-cycle explanations of residential mobility. Older individuals are less likely to move in a given year. Being married and having a child are also associated with lower rates of mobility, although these results are not statistically significant in the London-only sample. The odds ratios for the various types of rental housing are all greater than one in the full sample, meaning that individuals residing in these types of housing are more likely to move, relative to owners. For the full sample, the difference in mobility rates between council homes residents and owners is the smallest and is not statistically significant. For the London-only sample council tenants have lower rates of mobility than owners, as suggested in previous research (Cho and Whitehead, 2013). The categorical variables indicating the length of time since the individual last moved are all less than one, meaning that the later in a residence spell we observe an individual, the lower the probability they will move. Although the focus of this paper is not on modelling residential mobility per se, the intuitive appeal of the model enhances its credibility.

The second and third columns of Table 3 present the discrete time logistic regression results for the entire sample. The main independent variables of interest are residence in a gentrifying neighbourhood and the interaction between residence in a gentrifying neighbourhood and being low-income. If residence in a gentrifying neighbourhood (the reference category) is associated with a higher probability of mobility, we should observe an odds ratio that is statistically

Table 3. Multivariate discrete time logistic regression model.

	Full sample		London sample	
	Interaction with low income status	Interaction with working class status	Interaction with low income status	Interaction with working class status
Has a child	0.863 (4.42)**	0.892 (3.26)**	0.897 (0.85)	0.918 (0.64)
Age of survey respondent (reference category: aged 25–34)				
Aged 18–24	1.119 (2.61)**	1.221 (4.73)**	1.004 (0.03)	1.096 (0.60)
Aged 35–44	0.572 (12.14)**	0.569 (12.42)**	0.689 (2.28)*	0.683 (2.26)*
Aged 45–54	0.340 (18.06)**	0.346 (18.19)**	0.299 (5.25)**	0.305 (5.11)**
Aged 55–64	0.257 (18.84)**	0.273 (18.71)**	0.247 (5.27)**	0.257 (5.01)**
Aged 65 and above	0.184 (23.30)**	0.216 (21.27)**	0.229 (5.15)**	0.250 (4.83)**
Male	1.006 (0.20)	1.045 (1.52)	0.888 (1.18)	0.920 (0.80)
Marital status (reference category: married)				
Divorced, separated or widowed	1.452 (7.92)**	1.486 (8.56)**	1.258 (1.27)	1.223 (1.11)
Never married	1.105 (2.32)*	1.124 (2.67)**	0.992 (0.06)	0.959 (0.28)
UK citizen	1.181 (2.04)*	1.095 (1.15)	1.507 (2.30)*	1.431 (1.98)*
Type of tenure (reference category: owner)				
Council renter	1.044 (0.79)	1.063 (1.10)	0.605 (2.37)*	0.553 (2.49)*
Property company renter	3.444 (14.53)**	3.442 (13.99)**	1.631 (1.77)	1.541 (1.56)
Other association renter	1.215 (2.89)**	1.255 (3.28)**	1.710 (2.63)**	1.665 (2.42)*
Other renter	3.471 (30.72)**	3.522 (31.65)**	3.947 (10.77)**	3.604 (10.03)**
Resides in urban area	1.094 (1.89)	1.071 (1.48)		
Length of current residence spell (reference category: less than 3 years)				
Duration 3–5 years	0.851 (4.43)**	0.826 (5.28)**	0.801 (1.76)	0.776 (1.97)*
Duration 6–10 years	0.793 (4.98)**	0.779 (5.51)**	0.731 (1.81)	0.718 (1.91)
Duration greater than 10 years	0.589 (8.92)**	0.575 (9.85)**	0.558 (2.99)**	0.545 (3.05)**
Neighbourhood type (reference category: gentrifying neighbourhood)				
Disadvantaged non-gentrifying neighbourhood	1.016	0.958	3.175	1.895

(continued)

Table 3. (Continued)

	Full sample		London sample	
	Interaction with low income status	Interaction with working class status	Interaction with low income status	Interaction with working class status
Advantaged neighbourhood	(0.32) 1.034	(0.71) 0.931	(2.49)* 2.508	(1.40) 1.579
Low income	(0.66) 1.114	(1.47)	(2.07)* 3.085	(1.08)
Interaction between low income and disadvantaged non-gentrifying neighbourhood	(1.52) 0.982		(1.80) 0.096	
Interaction between low income and advantaged neighbourhood	(0.20) 0.875		(3.17)** 1.000	
Working class	(1.44)	0.894 (1.77) 0.919	0.294	0.407 (0.77) 1.635
Interaction between working class and disadvantaged non-gentrifying neighbourhood		(1.05) 0.998		(0.41) 2.384
Interaction between working class and advantaged neighbourhood		(0.03)		(0.74)
	84,107	76,743	6210	5688

Note:* $p < 0.05$; ** $p < 0.01$.

significant and less than one for residence in a disadvantaged, non-gentrifying neighbourhood. If low-income households are susceptible to higher rates of mobility in gentrifying neighbourhoods, we should observe an odds ratio that is statistically significant and less than one for the interaction between being low-income and residence in a non-gentrifying neighbourhood. Alas, neither of these patterns are in evidence. For the full sample, the odds ratio for non-

gentrifying disadvantaged neighbourhoods is greater than one, which would mean mobility was higher in these neighbourhoods than gentrifying neighbourhoods, but this result is not statistically significant. Moreover, the interaction terms between low-income status and residence in a non-gentrifying neighbourhood, and being working-class and residence in a non-gentrifying neighbourhood, are not statistically significant. For the full sample, there is

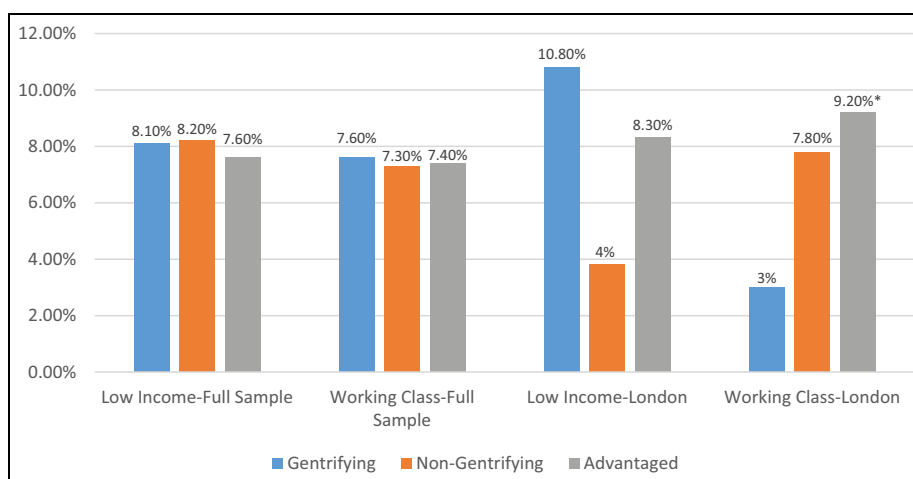


Figure 2. Predicted probability of moving.

Note: *Statistically significant difference between gentrifying and advantaged neighbourhoods.

no evidence that mobility rates are higher in gentrifying neighbourhoods or that low-income or working-class individuals are more susceptible to moving from gentrifying neighbourhoods.

The fourth and fifth columns of Table 3 present the results for the London-only sample. The odds ratio for the interaction between low-income status and residence in a non-gentrifying neighbourhood is statistically significant and substantially less than one. This means that low-income individuals in gentrifying neighbourhoods in London were more likely to move than similar individuals in non-gentrifying neighbourhoods. Working class Londoners, however, were no more likely to move if they lived in a gentrifying versus a non-gentrifying neighbourhood.

An alternative way of interpreting the interaction terms discussed above is presented in Figure 2, which shows the predicted probabilities of moving for poor and working-class individuals residing in gentrifying, non-gentrifying and advantaged neighbourhoods. In the full sample, the probability of moving varies little across the different types of neighbourhoods. In

London, however, the probability of moving is higher for low-income individuals in gentrifying neighbourhoods, and lower for working-class individuals in gentrifying neighbourhoods, than comparable individuals in non-gentrifying neighbourhoods. These differences, which were tested using a different statistical test than that used for the regression coefficients presented in Table 3, are not statistically significant. The only statistically significant difference in Figure 2 is for the predicted probability of moving between working class individuals in advantaged neighbourhoods in London and working class individuals in gentrifying neighbourhoods in London.

Overall, our results thus far provide mixed support for the thesis that gentrification leads to direct displacement. For the entire sample there is no evidence that direct displacement is associated with neighbourhoods that are gentrifying. In London, however, there is evidence consistent with the lowest-income individuals being displaced.

To adjudicate among the conflicting results, we estimate the relationship between gentrification and mobility using alternative

specifications. While the intuition behind gentrification leading to direct displacement is clear, as discussed above the precise functional form of this relationship is less certain. For example, we may be defining the gentry too broadly or demarcating gentrifying from non-gentrifying incorrectly. If patterns persist across alternative specifications this would inspire a degree of confidence in the results.

Robustness tests

The alternative specifications we attempted included:

1. Operationalising gentrification as the change in the absolute number of gentrifiers moving into a LLSOA. This approach relieves us of having to arbitrarily demarcate gentrifying from non-gentrifying LLSOAs.
2. Instead of defining gentrification as an increase in professionalisation at the 75th percentile or above for all LLSOAs, we used a more inclusive threshold of the 50th percentile or above.
3. We tested whether renters were especially susceptible to elevated rates of mobility in gentrifying neighbourhoods.
4. We defined the gentry more stringently to include only those with at least Level 4 (Level 4/5 in 2001 census) qualifications, that is, with a university degree, postgraduate degree or professional qualifications (e.g. accountant, nurse, teacher).

For all of these tests we estimated separate models for the entire sample and London. For robustness tests 1, 2 and 4 we estimated separate models for low-income and working-class individuals, respectively.

These alternative specifications did not produce any other statistically significant

results. Indeed, the finding that low-income individuals were more likely to move from gentrifying neighbourhoods in London did not prove robust across alternative specifications.

Discussion

This study tested whether quasi-experimental methods would detect direct displacement amidst residential turnover in gentrifying neighbourhoods in England and Wales, where mobility rates are lower than the US. Evidence of elevated mobility among low-income households in gentrifying neighbourhoods was found in only one specification and only for the London sample. The results presented here are thus for the most part inconsistent with the notion that gentrification leads to widespread direct displacement that manifests in higher mobility rates among residents of gentrifying neighbourhoods.

This result, and those from similar studies discussed above, seems to defy common sense. While this study has similar limitations to previous ones, as it utilises a survey that was not explicitly designed to measure displacement and must rely on indirect operationalisations of the key concepts involved, the above-mentioned critiques of this literature do not adequately explain why this study failed to detect elevated mobility rates out of gentrifying neighbourhoods. Our analysis compares gentrifying and non-gentrifying neighbourhoods that were similarly disadvantaged at the start of the study, and is therefore likely to capture the early stages of gentrification when many have posited displacement levels are highest. Due to the difficulty of operationalising the concepts involved, this study tested multiple specifications of gentrification and examined displacement rates for low-income and working-class individuals. There was also little evidence of *lower* mobility rates among

low-income households in gentrifying neighbourhoods versus non-gentrifying neighbourhoods, as Slater (2009: 305) and others assert is likely to be the case.

Yet anecdotal reports of direct displacement from gentrifying neighbourhoods abound. Documentaries and news articles have detailed the experiences of people struggling with rising housing costs in gentrifying neighbourhoods, including those identified in our study (Padilla, 2014). One tenant of a council estate targeted for regeneration in the East London borough of Hackney described how ‘when [he] realized how much his living costs would increase in the new flats, with a housing association as landlord rather than the council, he moved off Woodberry Down’ (Chakraborty and Robinson-Tillett, 2014). A Newham resident describes the Olympic regeneration legacy as follows:

If you are unemployed forget about being put on Newham’s housing register. I myself witnessed an elderly couple in retirement age being told that they would be moved out of the borough as there wasn’t accommodation for them in the borough! (Longlashes79, 2014)

How can we reconcile the quasi-experimental studies with first-hand accounts of gentrification-induced displacement?

Although existing critiques fail to provide a satisfactory explanation of this incongruence, the observations made by Slater (2009) and others that mobility rates among low-income households are relatively high is correct. Moreover, it is certain segments of the low-income – the young, private renters, those in unregulated housing – who have the highest mobility and are most vulnerable to displacement. For example, one report found that in Tower Hamlets, Newham and Hackney ‘a subset of households in the private rented sector or without secure accommodation are particularly highly mobile to

the point that they might be called transient’ (Scanlon et al., 2007). Finally, there is also some evidence that both low-income and non-low-income residents of poor neighbourhoods exit these neighbourhoods more quickly than other types of neighbourhoods, and that renters are more likely to stay put in neighbourhoods with rising rates of owner occupancy (Lee, 2014; Quillian, 1999, 2003). In societies where so much of what makes a neighbourhood desirable (e.g. low crime, good schools, shopping choices) is commodified or based on one’s ability to pay, people may be more likely to want to ‘stay put’ in neighbourhoods where the socioeconomic status and concomitantly desirability is increasing (Parkes and Kearns, 2003; Permentier et al., 2009).

Putting all of this together, one possible explanation for the incongruence between quasi-experimental results and anecdotal reports might be: A relatively poor neighbourhood that has experienced disinvestment will likely have above average mobility rates. This is because the poor move more frequently, renting (which is associated with higher mobility) is more common in poorer areas, and some residents will move to seek greener pastures. An influx of more affluent people and investment may occur to fill the vacuum, which is how gentrification occurs. Indeed, for the overall sample and for the London-only sample, a higher proportion of in-movers to gentrifying than non-gentrifying neighbourhoods had college degrees. This pattern was most marked in London, where 64% of in-movers to gentrifying neighbourhoods had college degrees, compared to 27% of in-movers to disadvantaged, non-gentrifying neighbourhoods.

Moreover, it may be the relatively high rates of turnover in some poorer neighbourhoods that allow gentrifiers to gain a foothold. As the neighbourhood gentrifies, some original residents who might otherwise have

left may decide to stay put because of improved amenities in the neighbourhood. Others who wish to leave may remain because of a lack of options elsewhere. Others may not like the changes and may move, and some will be directly displaced. However, the decrease in normal turnover rate and the increase of displacement counteract each other. From a view of 10,000 feet, therefore, the overall amount of residential turnover may appear to be little different from what was occurring prior to gentrification or what is occurring in non-gentrifying neighbourhoods. These findings should therefore not be interpreted as meaning that displacement does not occur in gentrifying neighbourhoods.

While quasi-experimental methods remain useful for examining gentrification and displacement on a broad scale, they do not give us a complete picture of the people, places and processes involved. There is thus a continuing need for in-depth studies of particular places to understand how neighbourhood change occurs and is experienced. On the ground, the challenge for those concerned about equitable outcomes in the face of gentrification is to identify the circumstances under which displacement occurs and to take steps to prevent it. Exclusionary displacement, or the loss of affordable units in gentrifying neighbourhoods, also remains a concern. If poorer households are not able to move into gentrifying neighbourhoods, where do they go? If the gentrifiers had not moved into the gentrifying neighbourhoods, where would they have moved? These and other questions insure that gentrification will continue to be a topic of much debate for the foreseeable future.

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Notes

1. Our analysis focuses exclusively on what Marcuse (1986) calls 'direct last-resident displacement', which occurs when the resident is forced to leave their current residence. We feel this type of displacement deserves the most scrutiny because it likely causes the most harm.
2. Thanks to one of the reviewers for suggesting this.

References

- Atkinson R (2000) Measuring gentrification and displacement in greater London. *Urban Studies* 37(1): 149–165.
- Brownhill S (1999) Turning the East End into the West End: The lessons and legacies of the London Docklands Development Corporation. In: Imrie R and Thomas H (eds) *British Urban Policy: An Evaluation of the Urban Development Corporations*. London, Sage Publications, pp. 41–64.
- Butler T and Hamnett C (2009) Walking backwards to the future: Waking up to class and gentrification in London. *Urban Policy and Research* 27(3): 217–228.
- Chakraborty A and Robinson-Tillett S (2014) The Truth about gentrification: Regeneration or Con trick? *Guardian*. Available at: <http://www.theguardian.com/society/2014/may/18/sp-truth-about-gentrification-how-woodberry-down-became-woodberry-park>.
- Cho Y and Whitehead C (2013) The immobility of social tenants: Is it true? Does it matter? *Journal of Housing and the Built Environment* 28(4): 705–726.
- David J and Thornley A (2012) Urban regeneration for the London 2012: Olympics: Issues of land acquisition and legacy. *City, Culture and Society* 1(2): 89–98.
- David Q, Janiak A and Wasmer E (2008) Local social capital and geographical mobility: A theory. IZA discussion paper no. 3668. The Institute for the Study of Labor, Bonn, Germany.
- Davidson M (2008) Spoiled mixture: Where does state-led 'positive' gentrification end? *Urban Studies* 45(12): 2385–2405.
- Davidson M and Lees L (2005) New-build 'gentrification' and London's riverside renaissance. *Environment and Planning A* 37(7): 1165–1190.

- Davidson M and Lees L (2010) New-build gentrification: Its histories, trajectories, and critical geographies. *Population Space and Place* 16(5): 395–411.
- DeVerteuil G (2011) Evidence of gentrification-induced displacement among social services in London and Los Angeles. *Urban Studies* 48(8): 1563–1580.
- DeVerteuil G (2012) Resisting gentrification-induced displacement: Advantages and disadvantages to staying put among non-profit social services in London and Los Angeles. *Area* 44(2): 208–216.
- Dieleman FM (2001) Modelling residential mobility: A review of recent trends in research. *Journal of Housing and the Built Environment* 16: 249–285.
- Ellen IG and O'Regan KM (2011) How low income neighborhoods change: Entry, exit, and enhancement. *Regional Science and Urban Economics* 41(2): 89–97.
- Fainstein S (2010) *The Just City*. Ithaca, NY: Cornell University Press.
- Freeman L (2005) Displacement or succession? Residential mobility in gentrifying neighborhoods. *Urban Affairs Review* 40(4): 463–491.
- Freeman L and Braconi F (2004) Gentrification and displacement: New York City in the 1990s. *Journal of the American Planning Association* 70(1): 39–52.
- Hamnett C (2003) Gentrification and the middle-class remaking of inner London, 1961–2001. *Urban Studies* 40(12): 2401–2426.
- Hamnett C (2009) Spatially displaced demand and the changing geography of house prices in London, 1995–2006. *Housing Studies* 24(3): 301–320.
- Hamnett C and Whitelegg D (2007) Loft conversion and gentrification in London: From industrial to postindustrial land use. *Environment and Planning A* 39(1): 106–124.
- Host Boroughs (2009) Strategic regeneration framework: An Olympic legacy for the host boroughs. Available at: <http://www.gamesmonitor.org.uk/files/strategic-regeneration-framework-report.pdf>.
- Institute for Social and Economic Research (ISER) (2014) British household panel survey. Available at: <https://www.iser.essex.ac.uk/bhps> (accessed 24 May 2014).
- Kearns A and Parkes A (2003) Living in and leaving poor neighbourhood conditions in England. *Housing Studies* 18(6): 827–851.
- Lee KO (2014) Why do renters stay in or leave certain neighborhoods? The role of neighborhood characteristics, housing tenure transitions, and race. *Journal of Regional Science* 54(5): 755–787.
- Leeser R (2010) English indices of deprivation 2010: A London perspective. Social Exclusion Team Intelligence Briefing. Greater London Authority, London, UK.
- Longlashes79 (2014) This is east London, not Park Lane. *Guardian*, 15 August. Available at: <http://www.theguardian.com/cities/2014/aug/15/stratford-regeneration-your-stories-olympic-legacy>.
- Marcuse P (1986) Abandonment, Gentrification and Displacement: The Linkages in New York City. In Smith N and Williams P (eds) *Gentrification of the City*. London: Unwin Hyman, pp. 153–177.
- Mayor of London (2007) Lower Lea Valley: Opportunity area planning framework. Strategic planning guidance. Greater London Authority and London Development Agency, London, UK.
- McKinnish T, Walsh R and White TK (2010) Who gentrifies low-income neighborhoods? *Journal of Urban Economics* 67(2): 180–193.
- Newman K and Wyly EK (2006) The right to stay put, revisited: Gentrification and resistance to displacement in New York City. *Urban Studies* 43(1): 23–57.
- Office for National Statistics (2014) Super output areas. Available at: <http://www.neighbourhood.statistics.gov.uk/dissemination/Info.do?page=nessgeography/superoutputaresexplained/output-areas-explained.htm> (accessed 23 May 2014).
- Owen DW and Green AE (1992) Migration patterns and trends. In: Champion AG and Fielding A (eds) *Migration Processes and Patterns, vol. 1: Research Progress and Prospects*. London, Bellhaven, pp. 17–38.
- Padilla AJ (2014) El Barrio tours. Available at: <http://elbarriotours.tumblr.com/>.
- Permentier M, van Ham M and Bolt G (2009) Neighbourhood reputation and the intention to leave the neighbourhood. *Environment and Planning A* 41: 2162–2180.

- Quillian L (1999) Migration patterns and the growth of high-poverty neighborhoods, 1970–1990. *American Journal of Sociology* 105(1): 1–37.
- Quillian L (2003) How long are exposures to poor neighborhoods? The long-term dynamics of entry and exit from poor neighborhoods. *Population Research and Policy Review* 22(3): 221–249.
- Rohe WM and Stewart LS (1996) Homeownership and neighborhood stability. *Housing Policy Debate* 7(1): 37–81.
- Rossi PH (1980) *Why Families Move*. Thousand Oaks, CA: SAGE Publications.
- Scanlon K, Travers T and Whitehead C (2007) Population churn and its impact on socioeconomic convergence in the five London 2012 host boroughs. Department for Communities and Local Government, London School of Economics, London, UK.
- Schachter J (2004) *Geographic Mobility 2002–2003*. Washington, DC: US Census Bureau.
- Seko M and Sumita K (2007) Effects of government policies on residential mobility in Japan: Income tax deduction system and the Rental Act. *Journal of Housing Economics* 16(2): 167–188.
- Slater T (2009) Missing Marcuse. *City* 13(2/3): 293–311.
- Taylor MF, Brice J, Buck N, et al. (2010) *British Household Panel Survey User Manual (Vol. A): Introduction, Technical Report and Appendices*. Colchester: University of Essex.
- Van Criekingen M (2009) Moving in/out of Brussels' historical core in the early 2000s: Migration and the effects of gentrification. *Urban Studies* 46(4): 825–848.
- Vigdor J (2002) Does gentrification harm the poor? *Brookings-Wharton Papers on Urban Affairs* 133–173.
- Watt P (2009) Housing stock transfers, regeneration and state-led gentrification in London. *Urban Policy and Research* 27(3): 229–242.
- Watt P (2013) It's not for us. *City* 17(1): 99–118.
- Wyly E, Newman K, Schafran A, et al. (2010) Displacing New York. *Environment and Planning A* 42(11): 2602–2623.